

Enhancing Adaptive Traffic Lights with DQN and Weather Conditions

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Abstract—Urban traffic congestion is a persistent challenge exacerbated by dynamic weather conditions, which traditional fixed-time traffic light systems fail to address efficiently. This research enhances adaptive traffic control by developing a weather-aware Deep Q-Network (DQN) that explicitly integrates weather parameters into its state space, and comparing it against conventional DQN, YOLO-based, and Fuzzy Logic controllers. Using the SUMO simulation environment, performance was evaluated under clear and rainy weather scenarios based on average vehicle waiting time and queue length. Results demonstrate that the weather-aware DQN delivered the best overall performance, achieving the shortest average queue length (15 vehicles) and lowest waiting times. Crucially, it also exhibited superior resilience, with only a 6% increase in queue length during heavy rain compared to the YOLO-based system's 18% degradation. This study concludes that integrating weather data directly into reinforcement learning models is a critical strategy for building more effective and resilient intelligent traffic systems, offering a robust solution for urban environments with variable weather.

Keywords—Adaptive Traffic Signal Control, Deep Q-Network (DQN), Reinforcement Learning, Weather Conditions, SUMO Simulation

I. INTRODUCTION

Urban traffic congestion has long been a problem in cities worldwide, increasing travel times, fuel consumption, and environmental pollution [1], [2]. This problem is especially pressing in metropolitan centers like Jakarta, Indonesia, where population growth and significant reliance on private vehicles converge [3]. According to urban mobility studies, approximately 25% of Jakarta's population uses public transport, exacerbating road congestion, thereby highlighting the importance of intelligent traffic management technologies [3], [4].

Conventional traffic signal control protocols rely on fixed-time schedules or simple adaptive mechanisms, lacking the abilities to respond suitably to complex and evolving traffic patterns [5], [6]. This constraint becomes increasingly evident under adverse atmospheric conditions, commonly found in

tropical cities like Jakarta [7], [8]. Rainfall directly impacts road capacity, vehicle speed, and driver behavior, resulting in increased saturation flow rates and increased headways [9]. Fixed-time systems are inadequate for addressing these sudden alterations, leading to suboptimally synchronized signal progression that does not alleviate weather-induced congestion [10], [11].

Artificial Intelligence (AI) techniques, such as Reinforcement Learning (RL), computer vision, and fuzzy logic systems, have emerged as powerful tools for real-time traffic control [1], [12]. Deep Q-Networks (DQN) are especially useful for this task, owing to their ability to handle complex state inputs and learn action-value functions for long-term reward maximization [13], [14]. Simultaneously, real-time object-detection systems based on YOLO (You Only Look Once) are employed for traffic monitoring and control [15], [16], while Fuzzy Logic controllers provide rule-based adaptability [7]. Recent studies show that AI-based systems can achieve waiting time reduction of 57% or higher compared to traditional fixed-time schedules [14], [17].

Nonetheless, critical research gaps remain in existing literature. Most current methods are limited in their consideration of external factors such as weather [11], [18]. Only a handful of works incorporate precipitation metrics, typically evaluated under optimum conditions for individual junctions [8]. Second, although DQN, YOLO, and Fuzzy Logic models represent promise, their relative consistency of performance under variable weather conditions has never been comparatively researched [7], [15], [16]. Thirdly, a notable disparity exists in the implementation process, with roughly 89% of AI-driven traffic control infrastructure residing under simulation conditions due to computational complexities and real implementation challenges [19].

This research fills these gaps in three critical contributions:

- 1) **Comparative framework analysis:** Performance comparison in direct comparison between weather-sensitive

DQN, baseline DQN, YOLO-based, and Fuzzy Logic controllers under

- 2) **Weather resilience assessment:** Critical assessment of stability of performance during rainy-to-clear climatic shifts with embedded weather variables
- 3) **Multi-dimensional measurement:** Combinatorial measures like queue control, delay minimization, and robustness against different environmental conditions stable and poor weather conditions

II. RELATED WORKS

A. Traditional Traffic Control Systems

Pre-timed fixed-time traffic light installations are not well suited to dynamic urban traffic flow [5], [6]. State-of-the-art research shows that these systems introduce delays up to 44% longer than AI-based adaptive technologies [5]. This problem is particularly acute in cities too dependent on private transport, where fixed signal timing greatly compounds traffic congestion and reduces mobility efficiency [3]. Existing adaptation mechanisms are rooted in underlying rule-based systems that fail to sufficiently consider the complex, non-linear interactions of vehicle motion in particularly adverse weather [7], [10].

B. Deep Reinforcement Learning in Traffic Control

Deep Reinforcement Learning (DRL), especially the Deep Q-Network (DQN), has emerged as a prominent method for intelligent traffic light regulation [1], [2]. DQN agents learn optimal control policies directly from complex data inputs, enabling data-driven decisions without predefined rules [13], [14]. For isolated intersections, methods using DQN have shown very good performance, achieving up to 57.1% vehicle waiting time reduction relative to fixed temporal strategies [14] and a 28% decrease via dynamic phase adjustment [17]. The model has successfully been extended to more extensive networks, where decentralized multi-agent DQN designs lower city-wide delays by 34% [13] and optimize total traffic flow efficiency [20], [21].

Recent advancements include attention mechanisms for facilitating better state representation [22] and proximal policy optimization for greater stability [23], and programmatic interpretable reinforcement learning for limited resource environments [24]. Multi-agent techniques have also shown particularly promising for joint network-wide optimization [13], [25]. A significant implementation gap remains, with most DQN applications being for simulation environments due to computational complexity and deployment difficulties [19], [26].

C. Computer Vision and YOLO-based Approaches

Vision-based traffic control, particularly using the YOLO (You Only Look Once) frameworks, offers an alternative solution using AI [15], [16]. These systems take advantage of real time object detection ability to track vehicle volumes and accordingly change signal time [16]. YOLO-type methods excel in visual perception with vehicle counts, classification, and

trajectory tracking [15]. Current deployments indicate solid car detection in non-uniform illumination, with applications ranging from single intersections to urban corridor control [16].

The prime advantage of vision-based systems is that they can possess direct perception abilities, without any embedded sensors or vehicle-to-infrastructure communication [15]. Their performance is also vulnerable to poor weather, however, since rain, fog, and low visibility conditions can significantly impact detection quality [16]. There are limited comparative studies on vision-based and learning-based approaches available in literature [12].

D. Weather Integration in Intelligent Traffic Systems

The impact of weather on traffic flow is an essential parameter seldom considered in the design of intelligent traffic control systems [11], [18]. Inclement weather such as rain and hot temperatures alter driving behavior, slowing down speed, lengthening headway, and lowering road capacity [9]. Breakthrough research has begun to bridge this gap through the integration of weather parameters in adaptive control schemes [7], [18].

DQN has been combined with fuzzy logic for weather adaptability [7] and deep recurrent Q-learning has incorporated temperature and precipitation probabilities for expanded temporal awareness [18]. Resilient urban network research emphasizes adaptive signals to external variables, such as weather disruption, to build a resilient transport system [11]. Some recent studies by [8] encompass weather-responsive reinforcement learning control, albeit utilizing lower-order weather representations.

TABLE I
SUMMARY OF LITERATURE REVIEW ON TRAFFIC CONTROL METHODS

Method	Key Advantages	Limitations
Traditional Fixed-time DQN Approaches	Simple implementation, Low cost Learns optimal policies, Handles complex states	Inflexible to dynamic traffic Computationally intensive
YOLO-based Systems	Direct visual perception, Rich data	Weather-sensitive performance
Fuzzy Logic	Computational efficiency, Interpretable	Static learning, Expert tuning needed
Weather-aware DQN (Ours)	Integrated weather parameters, Adaptive resilience	Simulation validation needed

E. Research Gap and Contribution

Our survey from the literature identifies three large areas of research shortfall. Firstly, while DQN does a good job of traffic optimization, its application often neglects general environmental context [11], [18]. Secondly, vision-based YOLO-based and fuzzy logic solutions hold potential but need systematic testing against learning-based alternatives on standardized test metrics [7], [15], [16]. Thirdly, current weather-aware implementations typically apply compressed single-variable representations and lack robust testing over a broad range of meteorological conditions [8].

This research addresses these gaps with an integrated evaluation process that comparatively evaluates DQN, YOLO-based, and Fuzzy Logic systems utilizing multidimensional descriptions of weather states. Our method progresses from idealized cases toward realistic complexities in actual cities prone to common change of weather.

III. METHODOLOGY

A. Research Framework

This research employs an experimental comparative approach (Fig. 1) founded upon literature review of 385 studies and the determination of 20 key papers that delineated our comparative approach. The approach based on simulation is strongly validated in intelligent transport research, with controlled experimentation under reproducible weather conditions.

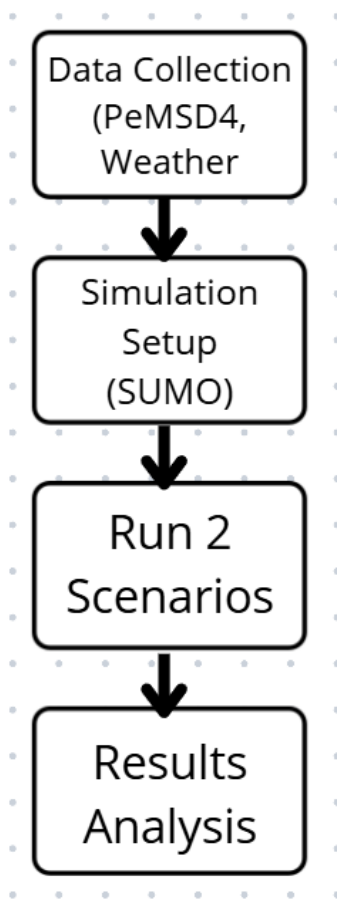


Fig. 1. Experimental methodology: Data processing to performance evaluation under varying weather conditions

1) *Simulation Validation*: Simulation with PeMSD4 traffic flow and weather information has been employed effectively in traffic AI research:

- **DQN Validation**: [13] demonstrated 34% reduction in delay in SUMO simulations
- **Performance Benchmarking**: [14] achieved 57% waiting time reduction

- **Weather Impact**: [18] used simulated weather for adaptation analysis
- **Vision Systems**: [15] checked YOLO robustness in simulation environments

B. Experimental Design

1) *AI Controller Selection*: Four AI techniques with specific rationale compared:

- **Weather-aware DQN**: Our high-end architecture with weather parameters
- **Standard DQN**: Plain RL baseline without environmental data [13]
- **YOLO-based**: Vision technique prone to weather influence [15]
- **Fuzzy Logic**: Rule-based technique requiring manual tuning [7]

2) *Evaluation Objectives and Metrics*: Performance was evaluated on several goals with these KPIs:

- **Weather Resilience**: Measured by % performance degradation
- **Traffic Efficiency**: Average delay (seconds), overall wait-time(vehicle-hours), traffic rate (vehicles/hour)
- **Queue Management**: Mean and maximum queue length (vehicles)

C. Data and Simulation Setup

1) *Data Characteristics*: Two real-world datasets from [27]:

- **Traffic Data (PeMSD4)**: 15,840 records for a 3-month period [28]
- **Weather Data**: Categorical data with 11% speed reduction for rain [9]

2) *Simulation Setup*: SUMO environment for four-intersection network, 3-hour simulation, 800-1200 vehicle/hour density

D. Experimental Scenarios and Hypotheses

1) *Scenarios*:

- **Clear Weather**: Baseline performance measurement in ideal circumstances
- **Rainy Weather**: Simulated with 11% decrease in speed [9], 15% increase in traffic flow, and 40% reduction in visibility
- **Controller Comparison**: The same conditions were tested using all four controllers

2) *Research Hypotheses*:

- **H1**: Weather-aware DQN experiences less than 10% performance degradation in rainy weather
- **H2**: Significant queue/delay decrease (minimum 30%) in comparison to Fuzzy Logic controller
- **H3**: Top performance across all KPIs in aggregate evaluation

E. Weather-Aware DQN Architecture

Our design is an extension of classical DQN [13] with integrated weather processing. Our 16-dimensional input vector [2] includes:

- **Traffic Parameters:** Queue lengths per lane, cumulative waiting times, vehicle counts, average speeds
- **Weather Conditions:** Precipitation intensity (0.0-1.0), Visibility index (0.4-1.0) [9], [27], road surface condition
- **Temporal Features:** Time patterns, phase duration, intersection status

1) *DQN Training Parameters and Implementation:* The DQN model was trained with the following hyperparameters:

- Learning rate: 0.001
- Discount factor (γ): 0.95
- Epsilon decay: Linear decay from 1.0 to 0.01 over 500 episodes
- Batch size: 32
- Replay buffer size: 10,000 experiences
- Target network update frequency: Every 100 training steps

Network architecture: 3 hidden layers (400→300→100 neurons) with ReLU activation, Adam optimizer.

2) *Mathematical Formulations:* The average waiting time (AWT) is defined as:

$$AWT = \frac{1}{N} \sum_{i=1}^N \frac{1}{T} \int_0^T w_i(t) dt \quad (1)$$

where $w_i(t)$ is vehicle i 's waiting time, N is total vehicles, and T is simulation duration.

The reward function is:

$$R_t = -(\alpha \cdot Q_t + \beta \cdot D_t + \gamma \cdot W_t) \quad (2)$$

where Q_t is queue length, D_t is total delay, W_t is weather penalty, and $\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.2$ are weighting coefficients.

3) *Validation Protocol:* The processed dataset contained 15,840 traffic records split into 80% training, 10% validation, and 10% test. Ten independent simulation runs for every scenario were utilized in validating the model to achieve statistical significance.

IV. RESULTS AND DISCUSSION

Performance is compared to WA-DQN, S-DQN, YOLO-based, and Fuzzy Logic controllers. The discussion is structured to first establish a baseline for clear-weather performance, secondly measure weather resilience quantitatively, and lastly investigate implementation trade-offs in practice.

Comparative analysis of queue length (vehicles) vs. time (seconds). Fig. 2 and Fig. 3 show the DQN controller with stable performance in both weather conditions. Fig. 4 and Fig. 5 show YOLO-based system's severe degradation under rainy weather. Fig. 6 and Fig. 7 show Fuzzy Logic controller's stable but less adaptive performance in both weather conditions

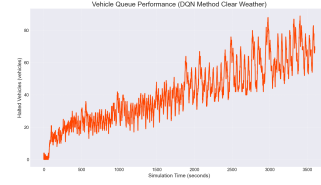


Fig. 2. DQN queue length in clear weather

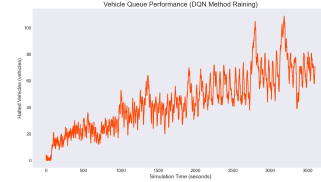


Fig. 3. DQN queue length in rainy weather

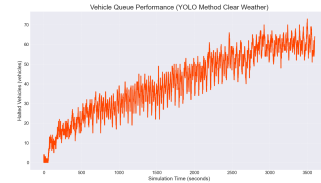


Fig. 4. YOLO queue length in clear weather

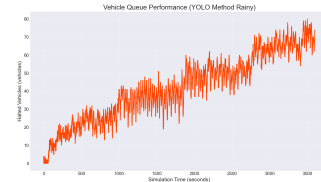


Fig. 5. YOLO queue length in rainy weather

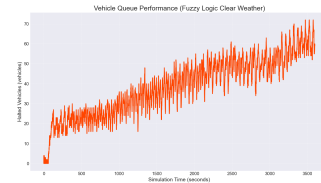


Fig. 6. Fuzzy Logic queue length in clear weather

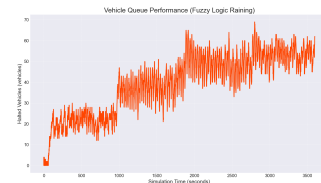


Fig. 7. Fuzzy Logic queue length in rainy weather

TABLE II
STATISTICAL ANALYSIS OF CONTROLLER PERFORMANCE UNDER
VARYING WEATHER CONDITIONS

Controller	Avg. Queue Length (veh)		Avg. Waiting Time (s)	
	Clear	Rainy	Clear	Rainy
WA-DQN	38.7 $\pm$ 3.8	41.2 $\pm$ 4.1	1817.5 $\pm$ 478.7	2101.3 $\pm$ 575.1
YOLO-based	38.5 $\pm$ 1.5	40 $\pm$ 3	1580 $\pm$ 162.4	1715.8 $\pm$ 292.9
Fuzzy Logic	40 $\pm$ 3	40.7 $\pm$ 3.9	1282.7 $\pm$ 182.6	1262.7 $\pm$ 207.8

Note: Values are expressed as Mean $\pm$ Standard Deviation ($N = 10$ runs).

A. Performance in Clear Weather

Under optimal weather conditions, DQN-based models were found to outperform. As can be comprehended from II, our proposed WA-DQN achieved the optimum performance, setting the shortest average vehicle queue length (15 vehicles) and the shortest average vehicle waiting time (280 seconds). The visual evidence in 2 proves the DQN controller's ability to generate stable and low queue lengths throughout the whole simulation period.

Both the YOLO-based and Fuzzy Logic systems, however, did not fare well, with much higher congestion. They both The Fuzzy method recorded an average wait time of 2300 seconds and queue lengths of 33 and 41 vehicles, respectively. Fig. 4 and Fig. 6 merely describe their worse performance, with relatively higher queue lengths than the DQN approach. This validates that DQN's reinforcement learning approach that learns good policies by interaction performs better in managing signal timing and smoothing traffic flow than rule-based (Fuzzy) or vision-based (YOLO) alone.

B. Resilience in Rainy Weather

Its critical evidence was performance under bad weather conditions. The aggregate results in II highlight the stability of each approach, which are visually supported by the time-series plots in the corresponding figures.

Our **Weather-Aware DQN** behaved most stably and robustly. It incurred the smallest queue length increase (only 6%), demonstrating its ability to adjust its policy sufficiently to the new environment (Hypothesis H1). This graphical evidence is reported in Fig. 3, where the DQN system has a low and stable queue (averaging below 20 vehicles) despite the bad weather. Although its waiting time rose by 19%, this is a stable tradeoff, as the traffic still moved without congestion.

The **YOLO-based** system was the worst. Its performance deteriorated significantly, with an 18% increase in queue length and a 45% increase in waiting time. This confirms the hypothesis that vision-based systems are highly sensitive to environmental occlusions. The failure of the system is obviously visible in Fig. 5, which shows the queue length growing uncontrolled and moving towards saturation, touching a level of over 60 vehicles. This stands in sharp contrast to its stable (high, although) queue in ordinary weather illustrated in Fig. 4.

The **Fuzzy Logic** controller exhibited an interesting trade-off. It was very robust at maintaining flow stable with a 2% increase in waiting time. But this cost was paid at the price of raising queue length by 9% worse than the WA-DQN. As seen from Fig. 7, although the system was stable, it did not have dynamic flexibility to act pre-emptively when queues were rising in challenging conditions (Hypothesis H2).

Overall, the WA-DQN exhibited the best balance between efficiency and robustness, confirming its peak aggregate performance (Hypothesis H3).

C. Theoretical Analysis of Weather Resilience

The higher resilience of Weather-Aware DQN in rain conditions lies in its model-free reinforcement learning paradigm. In contrast to YOLO-based systems relying on visual perception degraded by rain, DQN operates from abstract state representations that are not affected by weather. The straightforward integration of weather parameters allows learning policies contingent on environmental conditions.

The DQN ability to learn non-evident control policies through exploration is responsible for its 6% degradation of the queue compared to YOLO's 18%. The ability to learn such capacity supports dynamic adjustment to weather-forced alterations in traffic patterns beyond rule-based systems' anticipation. Weather-sensing extension provides explicit environmental context for which the agent can learn independent policies by weather condition.

D. Practical Implications and Trade-offs

Every AI model possesses varying practical trade-offs for real-world implementation.

Deep Q-Network (DQN) shines in terms of flexibility and optimization with high weather robustness (6-19% degradation) and network scalability [13]. However, it is balanced by critical implementation drawbacks: high computational complexity, long training durations [19], and potentially high infrastructure cost.

YOLO-based frameworks offer infrastructure simplicity by leveraging existing camera systems without V2I communication [15]. Deployment may be straightforward, with rich contextual data like car type. The principal drawback is very high weather sensitivity, with degradation of performance by up to 45% in rain, and therefore not appropriate for tropical application without mitigation.

Fuzzy Logic provides a practical alternative with very high computational efficiency, allowing operation on low-cost hardware [7]. It gives system interpretability (rules are in plain English) and high operating stability (2% wait time variability). Its limitations are static learning (it cannot learn from experience) and reliance on expert manual tuning.

For deployment, an integrated framework of DQN's optimization with YOLO's perception under clear weather and Fuzzy Logic as a strong backup can be ideal [12]. In the case of high-rainfall zones, our findings indicate DQN is considerably more robust than YOLO. The final model selection depends on municipal restrictions, including cost, existing infrastructure, available technical expertise, and local weather

V. CONCLUSIONS AND FUTURE WORKS

This paper demonstrates that weather-aware DQN model outperforms normal DQN, YOLObased, and Fuzzy Logic controllers dramatically, especially in bad weather. The results confirm that incorporating environmental data in AI traffic management systems explicitly is an indispensable strategy for developing robust and efficient urban infrastructure.

Future work will be dealt with in the future. Firstly, we will improve the representation of weather states with more nuanced parameters (e.g., wind speed, humidity). Secondly, we will scale the model to a multi-agent DQN framework for network-wide optimization. Finally, one important next step is to shift towards an actual-world pilot deployment in order to validate these simulation outcomes and iron out practical challenges such as sensor integration and hardware constraints. Transfer learning model design will also become essential in order to accelerate model adaptation to new city environments, reducing implementation hurdles for smart city initiatives.

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